Hydration Automation: A Smart Irrigation System for Urban Farming

Arnav Bharadwaj, Siddhi Kabadi, Inderpreet Kaur, Deepak Chowdary, Vamsi Krishna, Chandan Dhamande, Catherine Awad, Prashanth Gundala, Shruti Doshi
Faculty Advisor: Navid Shaghaghi · EPIC Research Lab & Frugal Innovation Hub @ Santa Clara University · EPICS in IEEE Funded · Partnering with Veggielution Urban Farm (San José, CA)

ABSTRACT

"Affordable Smart Irrigation for Every Urban Farm."

Urban farms face persistent challenges: excessive water consumption and high labor costs from manual irrigation. Commercially available smart-irrigation solutions are wired, proprietary, or priced beyond the reach of community farms serving low-income populations.

We present **Hydration Automation (HA),** an open-architecture, solar-powered, LoRa-connected precision irrigation platform for soil-based urban farms. HA couples low-cost custom sensing units (SUs) with a wireless base station (BS), cloud backend, and full-stack dashboard to enable autonomous, data-driven irrigation at a fraction of comparable commercial cost.

MOTIVATION & PROBLEM STATEMENT

Water waste: Schedule-based irrigation without real-time soil sensing causes systematic over-irrigation and nutrient leaching.

Labor burden: Manual monitoring demands dedicated staffing that under-resourced community farms cannot sustain across multiple growing beds.

Vendor lock-in: Incumbent systems (Rain Bird, Hunter, Tektelic) require AC power infrastructure, proprietary controllers, and paid cloud subscriptions.

Cost barrier: Commercial IoT irrigation nodes (METER TEROS 12, GroPoint, Tektelic Intellidrip) cost \$250–\$2,500+ per node: inaccessible to non-profit and community farms.

Research Goal: Deploy modular, solar-powered irrigation sensing under \$80/node with zero recurring software fees.

KEY COMPONENTS

Off-grid Sensing Unit (SU): PETG-encased unit with 4× capacitive soil moisture sensors, Adafruit Feather M0 (ATSAMD21, 48 MHz), solar + LiPo: fully field-deployable with no AC hookup.

Veggielution Community Farm

LoRa Sub-GHz LPWAN link: RFM9x radio supports multi-SU star topology to a single base station receiver with long-range, low-power wireless packets.

Wi-Fi Base Station (BS): Adafruit WINC1500 (802.11 b/g/n) aggregates LoRa packets and relays JSON records via SSL-encrypted HTTP POST to the cloud.

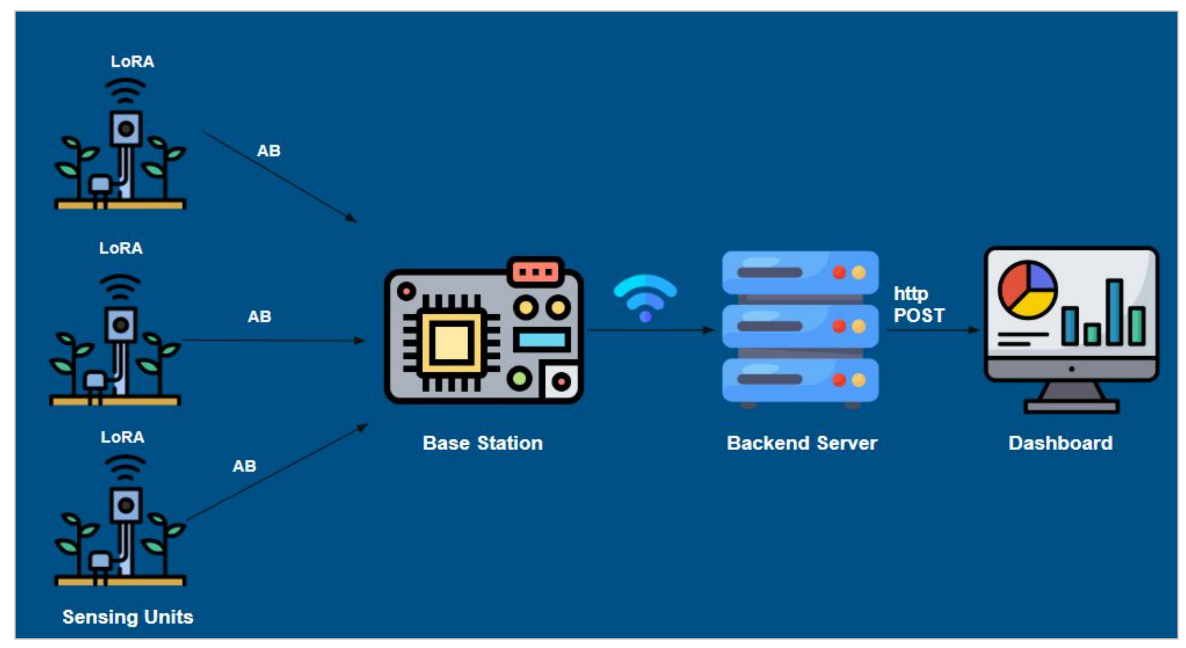
SU installed at the farm

Full-stack dashboard: Next.js/React + Express.js + MongoDB with Mapbox GL geolocation, Chart.js time-series, and Google OAuth 2.0 authentication.

Live deployment: 5+ SUs validated at **Veggielution** Community Farm, San José, CA in continuous field conditions.

SYSTEM ARCHITECTURE & DATA FLOW

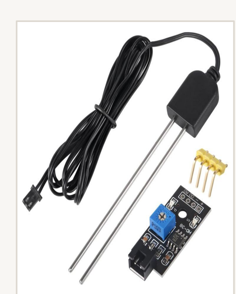
Multiple Sensing Units collect soil data and wirelessly send it to the Base Station via LoRa long-range, low-power radio. The Base Station securely uploads the data to a Backend Server over the internet, where it is displayed on a web dashboard for real-time monitoring and irrigation decisions.



SENSING UNIT HARDWARE

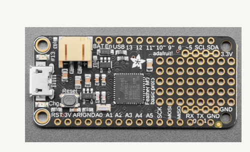
Soil Moisture Sensors

Capacitive probes measure VWC via dielectric permittivity. Up to 4 probes per SU for multi-depth or multi-bed profiling.



Adafruit Feather M0 + RFM9x

ATSAMD21 ARM Cortex-M0+ at 48 MHz. RFM9x LoRa radio handles sensor polling, duty-cycle sleep, and wireless transmission.



Solar + LiPo Power System

Solar panel trickle-charges 3.7 V LiPo via onboard USB charge controller. Enables indefinite off-grid deployment.

3D-Printed PETG Enclosure

Custom PETG housing provides UV resistance and IP-rated weather protection. Reproducible from open-source design files.

BASE STATION

The base station incorporates the **Adafruit WINC1500 Wi-Fi FeatherWing** stacked on the Feather M0 LoRa module. It continuously receives upstream LoRa packets from field SUs and forwards JSON records to the cloud backend via HTTPS.

Wi-Fi Connectivity

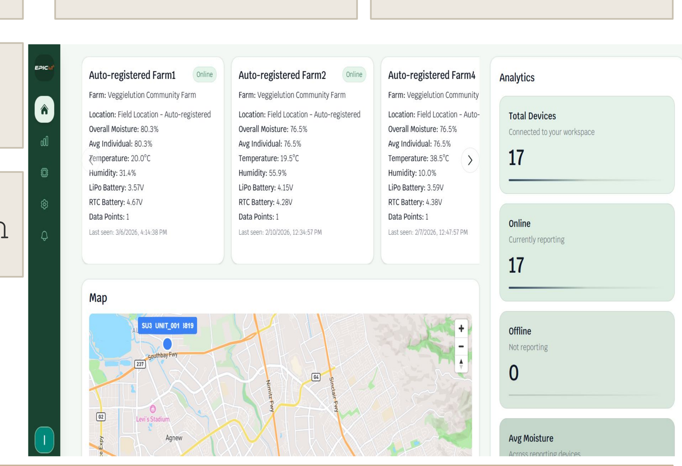
802.11 b/g/n (WINC1500). Supports WPA2-PSK for secured farm network access.

Secure Transmission

TLS-encrypted HTTP POST. Payload integrity verified from BS to cloud server.

DASHBOARD TECH STACK

FRONTEND		BACKEND	
Next.js	React	Express.js	MongoDB
TypeScript	Tailwind CSS	REST API	JSON/HTTPS
shadcn/ui	Mapbox GL		
Chart.js	Google OAuth		



Auth flow: User → Google Sign-In (OAuth 2.0) → Next.js/React UI → Express.js REST API → MongoDB (time-series JSON sensor records).

DEPLOYMENT METRICS

\$75	8+	4
TOTAL PER NODE	SUS DEPLOYED	SENSORS / UNIT

COMPETITIVE ANALYSIS

System	Price (USD)	Wireless	Solar	Modular
HA (ours)	~\$75/node	✔ LoRa	✔	✔
Rain Bird	\$500–\$700	Wired AC	✗	Controller
Hunter Irrig.	\$600–\$1,800+	Wired AC	✗	Controller
Tektelic Intellidrip	\$1,500–\$2,500+/yr	✔ LoRaWAN	✔	✔
Tevatronic	\$2000 - \$2400	✔ LoRaWAN	✔	✔

BILL OF MATERIALS (PER NODE)

Component	Approx. Cost
Adafruit Feather M0 + RFM9x LoRa radio	~\$34
Capacitive soil moisture sensors (×4)	~\$16
Mini solar panel + 3.7 V LiPo battery	~\$10
3D-printed PETG enclosure + fasteners	~\$2

CONCLUSIONS & FUTURE WORK

HA demonstrates that a fully autonomous, solar-powered, LoRa-connected precision irrigation platform can be deployed for under \$75/node without sacrificing wireless range, cloud connectivity, data security, or modularity. Live deployment at Veggielution validates the design under real agricultural conditions.

FUTURE DIRECTIONS

ML-based adaptive scheduling: Train models on historical VWC data + NOAA weather API to replace fixed thresholds.
Closed-loop valve actuation: Integrate solenoid valves with SU firmware for fully automated irrigation control.
Multi-farm deployment: Extend to multi-site installations; quantify water savings (L/day) vs. schedule-based baselines via A/B trials.
Open-source release: Publish CAD files, PCB schematics, and firmware under open-source license.

ACKNOWLEDGMENTS

This project was funded by **IEEE EPICS** and developed within the **EPIC Lab** and **Frugal Innovation Hub** at Santa Clara University. The team gratefully acknowledges **Veggielution** Community Farm for allowing for deployment.